THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Full Syllabus (Chapters 1–11) | Class 7 Science | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(d)** an acid	Acid.
2	**(d)** glow dimly	4.5 V is less than the rated 6 V.
3	**(a)** magnesium	Basic oxide, so metal.
4	**(d)** a reversible chemical change	Hydrated and anhydrous forms.
5	**(b)** growth rates vary and the cousin may have started his spurt	Individuals grow differently.
6	**(a)** A is a better conductor	Faster fall means faster conduction.
7	**(c)** 20 m/s	(150+250)/20.
8	**(b)** fat-rich food	Bile aids fat digestion.
9	**(b)** the covered half remains brown with iodine	Light needed.
10	**(c)** returns along the same path	For 90°, reversed.
11	**(c)** more extreme	Larger variation.
12	**(d)** 40 mL	Proportional.
13	**(b)** test with a tester; add salt to water to make it conduct	Dissolved salts supply ions.
14	**(d)** magnesium	Magnesium is highly reactive.
15	**(d)** formation of a new substance	New substance.
16	**(b)** larynx is growing and adjusting	Settles in time.
17	**(d)** reversed	Land breeze.
18	**(b)** 15 km	60 - 45.
19	**(d)** 1,15,200 times	80 x 60 x 24.
20	**(c)** light energy to chemical energy	Stored in glucose.

SECTION — B (Very Short Answer) — Answers — 6 × 2 = 12

Q.	Expected Answer / Value Points	Marks
21	Lemon juice is acidic (1); It neutralises the base so turmeric turns back to yellow (1).	1+1
22	Tungsten has a very high melting point (1); Copper would melt before glowing (1).	1+1
23	$Zn + H_2SO_4 \longrightarrow ZnSO_4 + H_2$ (1); Zinc sulphate and hydrogen (1).	1+1
24	Iron combines with oxygen and water (1); Rust has extra mass (1).	1+1
25	Hormones carry messages through blood (1); They control growth and mood (1).	1+1
26	Heating from the top gives little convection (1); From the bottom, hot liquid rises and cooler sinks (1).	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
27	Ratio 10:20 (1); 60 is three times (1); 30 mL acid (1).	1+1+1
28	6 x 1.5 = 9 V (1); Much higher than 3 V (1); Bulb may fuse (1).	1+1+1
29	Metal (1); Magnesium (1); Basic oxide (1).	1+1+1
30	A: no rust (no moisture) (1); B: no rust (no air) (1); C: rusts (air and water) (1).	1+1+1
31	Growth hormone lacking (1); Poor growth (1); Other glands underactive (1).	1+1+1
	OR: Thermocol best insulator (1); Ceramic next (1); Metal loses most heat (1).	1+1+1
32	Speed = 120/12 = 10 m/s (1); Distance in 30 s = 300 m (1); Platform = 300 - 120 = 180 m (1).	1+1+1
	OR: Fat digestion poor (1); Fewer enzymes (1); Special diet needed (1).	1+1+1

SECTION — D (Long Answer) — Answers

6 × 5 = 30

Q.	Expected Answer / Value Points	Marks
33	Dissolve equal tablets (1); Add to same volume of acid with indicator (1); Count amount needed to neutralise (1); Same conditions (1); Most effective needs the least acid change (1).	1+1+1+1+1
34	All bulbs go off (1); Only one path (1); Test each bulb with a tester (1); Replace the faulty one (1); Use a fresh bulb for check (1).	1+1+1+1+1
35	Colour and lustre (1); Conduction test (1); Magnet for iron (1); Burning gives acidic or basic oxide (1); Reaction with acid (1).	1+1+1+1+1
36	Set-ups: dry, no air, tap water, salt water (1); Same nails and period (1); Observe daily (1); Compare rust amounts (1); Salt water most, oiled none (1).	1+1+1+1+1
	OR: Poor energy and concentration (1); Disturbed sleep (1); Weight and mood problems (1); Plan: breakfast, sleep time, screen limits (1); Monitor progress (1).	1+1+1+1+1
37	Same volume and initial temperature (1); Wrap each beaker (1); Measure every 5 min (1); Same room conditions (1); Wool keeps heat longest (1).	1+1+1+1+1
	OR: 0-10 s: 2 m/s (1); 10-15 s: rest (1); 15-20 s: 6 m/s (1); Average 50/20 = 2.5 m/s (1); Graph has a flat part (1).	1+1+1+1+1
38	Rest: count breaths per minute (1); After walking (1); After running (1); Table of results (1); Rate rises with activity (1).	1+1+1+1+1
	OR: Vary lamp distance (1); Count bubbles per minute (1); Keep temperature and CO2 same (1); Table and graph (1); Rate rises with light (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERPIECE

THE MASTERMIND COACHING CLASSES

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ANSWER KEY & MARKING SCHEME SET 2

Full Syllabus (Chapters 1–11) | Class 7 Science | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(c)** umbra	Complete shadow.
2	**(b)** more frequent leap years	Larger quarter.
3	**(d)** acidic	Acid rain.
4	**(c)** a conductor	Current passes through it.
5	**(b)** zinc	Acid reacts.
6	**(a)** store in dry place or oil them	Keeps moisture away.
7	**(b)** a rapid rise during the growth spurt	Sudden increase.
8	**(a)** sea breeze	Fluid motion.
9	**(d)** 1.82 s	Mean 18.2 / 10.
10	**(d)** to provide a large surface area for absorption	Large area.
11	**(a)** nitrogen-fixing bacteria	Rhizobium.
12	**(c)** virtual image position follows the law of reflection	Geometric consequence.
13	**(d)** the Earth always has a shadow but the Moon enters it only during eclipses	Shadow is always there.
14	**(d)** an acid	An acid neutralises the base.
15	**(b)** glow very bright and may fuse	Exceeds rating.
16	**(b)** sulphur	Non-metal.
17	**(a)** copper sulphate	Copper deposits.
18	**(c)** it is within the normal range	Normal variation.
19	**(a)** it absorbs more radiation	Dark absorbs.
20	**(d)** 300 m	300/30.

SECTION — B (Very Short Answer) — Answers — 6 × 2 = 12

Q.	Expected Answer / Value Points	Marks
21	Distance 500 m in 30 s (1); 500/30 = 16.67 m/s, i.e. 60 km/h (1).	1+1
22	Respiration releases energy as heat (1); Metabolism increases after a meal (1).	1+1
23	The plant uses CO2 produced and gives oxygen (1); The candle can burn longer (1).	1+1
24	Angle of incidence is 0° (1); The ray retraces its path (1).	1+1
25	Moon reflects only a small fraction (1); Light is weak (1).	1+1
26	Use litmus or turmeric (1); Different colours appear (1).	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
27	Hydrilla in water under funnel (1); Test tube collects gas (1); Control in dark (1).	1+1+1
28	i = r = 55° (1); Angle between rays 110° (1); Angle with mirror 35° (1).	1+1+1
29	15° in 1 hour (1); 1° in 4 minutes (1); 15° takes 1 hour (1).	1+1+1
30	Some are more sensitive (1); Colour changes differ (1); Choose according to need (1).	1+1+1
31	They need a large current (1); Thin wiring may overheat (1); Risk of fire (1).	1+1+1
	OR: $MgO + H_2O \longrightarrow Mg(OH)_2$ (1); $SO_2 + H_2O \longrightarrow H_2SO_3$ (1); Basic and acidic (1).	1+1+1
32	Salt and sand do not react (1); Iron and sulphur react (1); New compound formed (1).	1+1+1
	OR: Avoid excess sweets (1); Carry medicine (1); Eat at regular times (1).	1+1+1

SECTION — D (Long Answer) — Answers

6 × 5 = 30

Q.	Expected Answer / Value Points	Marks
33	Each mirror forms one image (1); That gives two images (1); The image in one mirror acts as an object for the other (1); It forms a third image (1); Total 3 images (1).	1+1+1+1+1
34	Fix a stick on the globe (1); Place lamp at the Sun's position (1); Tilt the globe (1); Measure shadows at two latitudes (1); Shadow longer at higher latitudes (1).	1+1+1+1+1
35	Some crops prefer acidic soil (1); Others basic (1); Lime reduces acidity (1); Manure improves soil (1); Test regularly (1).	1+1+1+1+1
36	Marks show quality (1); Safer products (1); Fuse rating matches use (1); Do not use higher rating (1); Check before buying (1).	1+1+1+1+1
	OR: Burn Mg ribbon (1); Dissolve ash in water (1); Burn sulphur and dissolve gas (1); Test both with litmus (1); Compare (1).	1+1+1+1+1
37	Burn a candle in a closed jar (1); It goes out (1); Add lime water (1); It turns milky (1); Conclude (1).	1+1+1+1+1
	OR: Respect for others (1); Say no to unwanted behaviour (1); Report abuse (1); Trusted adults (1); Laws protect (1).	1+1+1+1+1
38	Black in sun highest (1); White in sun lower (1); Both in shade low (1); Differences show absorption (1); Conclude (1).	1+1+1+1+1
	OR: Distance 80 km in 2 h (1); Average 40 km/h (1); Speedometer shows instantaneous (1); Values differ (1); Both useful (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERKEY

ANSWER KEY & MARKING SCHEME SET 3

Full Syllabus (Chapters 1–11) | Class 7 Science | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(b)** digestion of all three food types	Pancreatic juice is important.
2	**(a)** chlorophyll is needed	Pigment.
3	**(d)** increases by 15°	Twice the turn of the mirror (10 x 2).
4	**(c)** 12 hours	Half.
5	**(d)** 30 mL	Doubling.
6	**(a)** tap water has dissolved salts	Dissolved ions conduct.
7	**(c)** Fe	Iron.
8	**(a)** lime water test	Test gas.
9	**(d)** menstruation	Ovary hormones needed.
10	**(a)** nearly stops	Temperatures equalise.
11	**(c)** 360 km	Gap 120 km, closing 30 km/h in 4 h.
12	**(b)** 8000 mL	16 x 500.
13	**(a)** chemical energy to usable energy	Released.
14	**(a)** shadow	Dark.
15	**(a)** the Earth moves around the Sun meanwhile	Extra two days to realign.
16	**(c)** it is a base used to relieve acidity	Carbonates react with acids.
17	**(d)** an insulator	Current does not pass.
18	**(c)** copper	Good conductor.
19	**(d)** apply oil and keep dry	Oiling.
20	**(c)** around 11 to 13 years	Growth spurt.

SECTION — B (Very Short Answer) — Answers — 6 × 2 = 12

Q.	Expected Answer / Value Points	Marks
21	Metals are good conductors (1); They carry current to the appliance (1).	1+1
22	They dissolve in rain to form acids (1); Such as SO_2 giving sulphurous acid (1).	1+1
23	Burning forms ash and gases (1); Tearing only changes size and shape (1).	1+1
24	Testosterone promotes muscle growth (1); Greater rise in boys (1).	1+1
25	Evaporation takes heat (1); Keeps water cool (1).	1+1
26	Time spent at each speed differs (1); Average speed = total distance/total time (1).	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
27	Steel conducts (1); Plastic does not (1); Steel handle becomes hot (1).	1+1+1
28	Time = 3 h (1); Distance = speed x time (1); 40 x 3 = 120 km (1).	1+1+1
29	Move to site (1); Engulf germs (1); Defence (1).	1+1+1
30	Light is available at the top (1); Competition for sunlight (1); More photosynthesis (1).	1+1+1
31	Fix a stick vertically (1); Mark shadow lengths (1); Shortest shadow is at noon (1).	1+1+1
	OR: Earth's shadow is bigger (1); Moon's shadow small (1); Speeds (1).	1+1+1
32	$H_2SO_4 + 2\,KOH \longrightarrow K_2SO_4 + 2\,H_2O$ (1); Acid and base (1); Salt and water (1).	1+1+1
	OR: Switch off the supply (1); Dry hands (1); Use proper bulb (1).	1+1+1

SECTION — D (Long Answer) — Answers — 6 × 5 = 30

Q.	Expected Answer / Value Points	Marks

33	Lung damage (1); Blood vessel damage (1); Heart disease (1); Toxins strain kidneys (1); Avoid smoking (1).	1+1+1+1+1
34	Set up three conditions (1); Same plant type (1); Starch test after some days (1); Only suitable one forms starch (1); Conclusions (1).	1+1+1+1+1
35	Pinhole real inverted (1); Mirror virtual erect (1); Pinhole needs a screen (1); Mirror image behind (1); Both use straight-line light (1).	1+1+1+1+1
36	Regular motions (1); Records of old eclipses (1); Calculations (1); Computer models (1); Accurate predictions (1).	1+1+1+1+1
	OR: Soil treatment (1); Antacids (1); Acid rain (1); Treating wastes (1); Principle: acid + base gives salt + water (1).	1+1+1+1+1
37	Switch off unused devices (1); Use LED bulbs (1); Use daylight (1); Do not overload (1); Service appliances (1).	1+1+1+1+1
	OR: $2\,Mg + O_2 \longrightarrow 2\,MgO$ (1); $Mg + 2\,HCl \longrightarrow MgCl_2 + H_2$ (1); $Mg + CuSO_4 \longrightarrow MgSO_4 + Cu$ (1); Mg is reactive (1); Products observed (1).	1+1+1+1+1
38	LPG burns: chemical (1); Milk warms: physical (1); Milk boils: physical (1); Milk burns at bottom: chemical (1); Black deposit is carbon (1).	1+1+1+1+1
	OR: Nerves: fast signals (1); Hormones: slower, lasting (1); Adrenaline for stress (1); Sex hormones for puberty (1); Both coordinate (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND